

Success Metrics (KPIs) — counts-only, no-flow-by-construction

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Document role. This is the Volume 1 (Product & Requirements) nano-detail spec defining HelixVPN's product **KPIs** and the **telemetry that derives them** — strictly **counts and aggregates only**. The defining constraint of this document is that it is bound by `v05-security/no-logging-as-code.md`: a KPI we *cannot* measure without a per-user flow, a destination, or a timestamp-of-use is a KPI we **deliberately do not have** — and that absence is a *product feature*, not a gap. Every KPI names its counts-only metric source (citing `v03-control-plane/svc-telemetry.md`) and the privacy boundary it respects.

Document is a SPEC. It states *what we measure, from which counts-only series, against which target, within which privacy boundary*; it does not build the product. Every numeric target is a TARGET marked UNVERIFIED per §11.4.6 — a product KPI threshold is a business target, not a measured fact, until the instrumented series produces data.

Evidence base. Cross-references the Volume 1 overview (principles P7 no-logging, parity F14; differentiators X1–X5; the MVP DoD §10.2), the telemetry counts-only metric registry and its forbidden-label cardinality bound (`v03-control-plane/svc-telemetry.md` §3), the coordinator convergence SLO (`v03-control-plane/svc-coordinator.md` §7), and the no-logging invariant (`v05-security/no-logging-as-code.md` §2 the stored-vs-never-stored partition).

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1. The measurement charter (the privacy boundary is the design)

Most products measure success with per-user funnels, destination analytics, and session timelines. HelixVPN **cannot** — by construction (P7 / S6): no durable connection/traffic/flow/session table exists, the event bus carries no src/dst/bytes/dns field, and /metrics forbids any tenant_id/device_id/ip label. Every KPI in this document is therefore derived from **aggregate counters, SLO histograms, and bounded gauges** — never from a per-user trail.

The charter, stated as three rules every KPI obeys:

1. **Counts and aggregates only.** A KPI's source is a Prometheus counter, histogram, or bounded gauge from the telemetry registry (svc-telemetry.md §3), or an aggregate roll-up of those — never a per-user row (there is none).
2. **No forbidden label.** A KPI may slice by a *bounded* label (action, event_type, stream, build version) but **never** by tenant_id, device_id, overlay_ip, or any traffic/PII-shaped key — that label set is forbidden by the cardinality guard (svc-telemetry.md §3.1).
3. **The absence is a feature.** Where a conventional KPI would need a per-user flow / destination / timestamp-of-use, HelixVPN has no such KPI **on purpose** (§8). "We can't tell you which sites a user visited or when they were online" is the product promise, surfaced as a measurement boundary.

flowchart TD

```

SRC[telemetry registry · counts-only series] --> KPI[product KPIs]
  GUARD{forbidden label?<br/>tenant_id/device_id/ip} -- yes --> REJECT[NOT a KPI · privacy boundary]
  GUARD -- no --> KPI
    NOFLOW[no durable connection/flow/session table] --> ABSENT[deliberately-absent KPIs §8]
    ABSENT --> FEATURE[the absence IS the product promise P7]
    KPI --> TARGET[each KPI: definition · source · TARGET UNVERIFIED · boundary]
```

§11.4.6 honesty. Every numeric target below is a *business TARGET* marked UNVERIFIED — KPI thresholds are goals, not measurements. A KPI is "met" only when its named counts-only series produces data showing it; the threshold written here is never asserted as already achieved.

2. What we can measure vs what we deliberately cannot

The closed partition, drawn directly from the no-logging stored-vs-never-stored model (no-logging-as-code.md §2) projected onto the KPI layer:

Measurable (counts/aggregates exist)	Deliberately NOT measurable (no data exists)
Total devices enrolled (registry count)	Which destinations any device reached
Devices currently online (bounded presence gauge, aggregate)	When a specific user was online (no session timeline; <code>last_seen_at</code> coarse ≥ 5 min)
Total connectors / advertised networks (topology count)	Which user used which network at which time
Handshakes total, by transport-kind (aggregate counter)	Per-user transport history / per-user flow
Convergence/revoke SLO percentiles (histograms)	Per-user latency or per-destination latency
Aggregate bytes-total (no per-peer, no per-user)	Per-user / per-destination byte counts
Audit control-action counts, by action (bounded)	Any traffic event (none is audited — none exists)
Ladder escalation outcomes, aggregate by rung	Which user escalated on which network (no SSID, no endpoint, I5)

The right column is not "unimplemented" — it is **architecturally impossible** to measure because the data is never written (S6). This is the §11.4.112 structural property turned into a product virtue: the metric cannot be reconstructed because there is nothing to reconstruct it from.

3. Adoption & reach KPIs (counts-only)

KPI	Definition	Counts-only source	Target
K-ADOPT-1 Devices enrolled	total devices ever enrolled per tenant (and global aggregate)	registry row count; <code>device.enrolled</code> audit counter (<code>helix_audit_events_total{action="device.enrolled"}</code>)	TARCO growth curve - UNVER

KPI	Definition	Counts-only source	Target
K-ADOPT-2 Active devices	devices currently online	helix_presence_online_devices (bounded gauge, reconciled from SCARD)	TARGET — UNVERIFIED
K-ADOPT-3 Networks joined	total connectors attached + advertised prefixes (the X1 multi-network differentiator)	registry/topology counts (advertised_prefixes rows)	TARGET — UNVERIFIED
K-ADOPT-4 Multi-network reach	distribution of networks-reachable-per-user (the 1 user → N nets headline)	derived aggregate from compiled-policy VisibleTo cardinality (counts, not identities)	TARGET — UNVERIFIED
K-ADOPT-5 Apps in use	which of the three app classes (Access/Connector/Console) are active	helix_build_info + open-stream class aggregate	TARGET — UNVERIFIED

K-ADOPT-2 boundary (§11.4.6). "Active devices" is an *instantaneous* aggregate count derived from the ephemeral presence set — it deliberately cannot answer "was device X online yesterday at 14:00" because presence is TTL'd and `last_seen_at` is coarsened to ≥ 5 min and carries no destination (no-logging §2.2). The point-in-time count is the metric; the timeline is the deliberately-absent one (§8).

4. Reliability & performance KPIs (SLO-derived)

These KPIs are roll-ups of the SLO histograms the coordinator + telemetry already expose (svc-coordinator.md §7, svc-telemetry.md §3.2). They measure *system* quality, never *user* behaviour.

KPI	Definition	Counts-only source	Target
K-REL-1 Convergence SLO attainment	% of time p99(helix_reconcile_seconds) < 1 s	helix_reconcile_seconds histogram	TARGET p99 < 1 s (DoD-5 UNVERI
K-REL-2 Revoke SLO attainment	% of revokes enforced < 1 s	helix_revoke_enforce_seconds histogram	TARGET 1 s — UNVERI
K-REL-3 Availability (fail-static)	tunnel-drops attributable to control-plane outages	chaos-test outcome + helix_presence_backend_up gauge	TARGET drops of control- plane or (NFR-2
K-REL-4 Event-pipeline health	DLQ accumulation + stream backlog	helix_events_dlq_total, helix_stream_pending_entries	TARGET DLQ fla backlog bounded UNVERI
K-REL-5 No-leak resource posture	coordinator memory slope over soak	process_resident_memory_bytes deriv	TARGET slope ≈ over 24 (NFR-1
K-REL-6 Handshake success rate	successful WG handshakes / attempts (aggregate)	aggregate handshake counter (edge / metrics)	TARGET — UNVERI

K-REL-1 is the product's flagship quality KPI. It is the parity-grade real-time promise (P4) made into a falsifiable series. It carries **no** tenant or device label (§3.1 cardinality guard), so it measures the *system's* convergence, never any user's activity.

5. Privacy-integrity KPIs (the guarantee is itself a KPI)

The most important KPIs invert the usual relationship: the *strength of the privacy guarantee* is measured, and any erosion is a release-blocking failure.

KPI	Definition	Counts-only source	Target	Privacy boundary
K-PRIV-1 No-logging gate green	schema-lint exit-0 against migrations AND the deployed DB	schema lint exit code (pre-build + post-deploy runtime signature)	TARGET 100 % green (NFR-300/308)	meta-only; the gate proves <i>absence</i> , stores nothing
K-PRIV-2 Lint not a tautology	paired §1.1 mutation: planted flows table → lint FAIL	meta-test result	TARGET golden-bad FAILs every build (NFR-301)	meta-only
K-PRIV-3 Bus payload clean	zero traffic-shaped field in any event payload	payload-lint result	TARGET 0 forbidden fields (NFR-302)	meta-only
K-PRIV-4 Audit control-only	audit rows are control actions only; meta-shape guard active	helix_audit_events_total{action} (closed vocab) + reject counter	TARGET 0 traffic-shaped meta rejected-into-store (NFR-303)	bounded action label only
K-PRIV-5 Metric cardinality bound	every collector's label set $\subseteq$ allow-list	label-audit unit test	TARGET 100 % pass (NFR-306)	the KPI <i>enforces</i> the boundary it lives under
K-PRIV-6 Anonymous-enroll availability	tenants can enroll devices with no email/SSO	integration test result	TARGET available (NFR-307, F15)	the feature that makes the boundary user-meaningful

K-PRIV-5 is reflexive. It measures that the KPI layer itself never grew a forbidden label — the success metric for "we did not accidentally start tracking users" is itself a counts-only, label-bounded check. A green K-PRIV-1..6 is the captured-evidence proof (§11.4.69) that the privacy promise is mechanically intact, not merely asserted.

6. Censorship-resistance KPIs (aggregate ladder outcomes)

The transport ladder records **aggregate, no-per-user** escalation outcomes (I5) — exactly enough to know "is obfuscation working in aggregate" without ever knowing *who* escalated *where*.

KPI	Definition	Counts-only source	Target	Privacy boundary
K-CENS-1 Obfuscation success rate	% of connect attempts that reach a working rung (aggregate)	aggregate ladder-outcome counter by terminal rung	TARGET high — UNVERIFIED	aggregate by rung-kind; no SSID, no endpoint, no per-user (I5)
K-CENS-2 DPI-survival throughput	obfuscated throughput as % of plain (the Phase-0 G2 bar)	benchmark capture (netns DPI rig)	TARGET $\geq$ 50 % of plain (NFR-002) — UNVERIFIED until G2	benchmark, not production user data
K-CENS-3 Escalation-depth distribution	how far up the ladder attempts go, in aggregate	histogram of terminal-rung index (counts)	TARGET — UNVERIFIED	aggregate distribution; no per-network identity (fingerprint hash only)
K-CENS-4 Time-to-working-rung	aggregate latency to a successful WG handshake under escalation	aggregate timing histogram	TARGET bounded (NFR-006) — UNVERIFIED	aggregate timing; no per-user trace

K-CENS boundary (§11.4.6). The ladder's per-network memory persists a fingerprint *hash* + rung index — so K-CENS-3 can report "X% of networks needed MASQUE" in aggregate but **cannot** name a network, an SSID, or a user. The escalation event trace is the captured evidence (L-I9) and carries no destination.

7. Self-host adoption KPIs (the differentiator)

Self-hostability (P6 / X3) is the product's reason to exist; its success is measured by deployment-shape counts, never by reaching into deployed instances (a self-hoster's instance is theirs — HelixVPN cannot and must not phone home per-user data).

KPI	Definition	Counts-only source	Target	Privacy boundary
K-SELF-1 Self-host bootstrap success	a clean VPS reaches a working deployment via <code>helixvpnctl init</code> (DoD-1)	bootstrap test / Challenge outcome	TARGET 100 % on the supported path — UNVERIFIED	test outcome; no telemetry from operator instances by default
K-SELF-2 8-criteria DoD attainment	how many of the MVP DoD criteria pass on a clean self-host	DoD acceptance-suite result (8 criteria)	TARGET 8/8 — UNVERIFIED	acceptance evidence, not user data
K-SELF-3 Single-pod resource fit	the self-host pod runs within homelab resource bounds + the §12.6 60 % ceiling	host resource sampler (§11.4.24)	TARGET within budget — UNVERIFIED	host-side metric on the <i>test</i> host, not operator instances
K-SELF-4 Image-set parity	the same OCI image set serves self-host and fleet (no code fork)	build/artifact check	TARGET one image set — UNVERIFIED	artifact metadata, not user data

No phone-home by default (§11.4.6 / P7). A self-hosted instance does not emit per-user telemetry to any HelixVPN-operated endpoint; K-SELF KPIs are measured from the *project's own* acceptance suite and test hosts, never by harvesting operator deployments. A managed SKU may collect *aggregate* operational counters from its *own* fleet under the same counts-only boundary — never per-user flows.

8. The deliberately-absent metrics (and why each absence is a feature)

This is the document's thesis table: the conventional KPIs HelixVPN **deliberately does not have**, each paired with the product promise its absence encodes.

Conventional KPI we do NOT have	Why it cannot exist	The feature the absence encodes
Destinations per user / "top sites"	no durable destination column anywhere (S6)	"we never know where you go" — the core privacy promise
Per-user session timeline / "usage hours"	presence is ephemeral TTL; <code>last_seen_at</code> coarse ≥ 5 min, no destination	"we can't reconstruct when you were online"
Per-user bandwidth / data-cap analytics	no per-user/per-peer byte counts; only aggregate bytes-total	"we don't meter or profile your traffic"

Conventional KPI we do NOT have	Why it cannot exist	The feature the absence encodes
Per-user transport history	ladder memory is a fingerprint hash + rung index, no per-user link (I5)	"we don't track which networks you fought censorship on"
DNS query analytics / "popular domains"	no DNS query log table (S6)	"we never log your DNS" (F13)
Funnel/cohort analysis keyed to a person	no tenant_id/device_id label on metrics; no PII-linked event	"you can be fully anonymous (F15)"
Per-tenant breakdown on public /metrics	forbidden labels; per-tenant only behind authenticated / v1/stats	"your population is not leaked to a scraper"

§11.4.112 framing. Each absence is a *structural impossibility*, not a backlog item: the data is never written, so the metric is unreconstructable. Reintroducing any of these would require adding a forbidden table/column/label that the schema-lint + payload-lint + cardinality guard would *fail the build* over (K-PRIV-1..5). The absence is therefore self-enforcing.

9. KPI → metric source → privacy-boundary traceability

KPI family	Counts-only source	Principle / parity	Privacy guard that bounds it
Adoption (K-ADOPT)	registry counts, presence gauge, audit counters	X1 multi-network	no tenant_id/device_id label; per-tenant only behind /v1/stats
Reliability (K-REL)	SLO histograms, process gauges	P4 push, P1 fail-static	no per-user/tenant label on histograms
Privacy-integrity (K-PRIV)	lint exit codes, label-audit tests	P7 no-logging, F15	the KPIs <i>are</i> the boundary enforcers
Censorship (K-CENS)	aggregate ladder-outcome counters	F2/F6/F7	fingerprint hash only; no SSID/endpoint/user (I5)
Self-host (K-SELF)	acceptance-suite + build checks	P6/X3	no phone-home; test-host metrics only

flowchart LR

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subgraph Sources[counts-only telemetry]
    C1[registry counts]
    C2[presence gauge aggregate]
    C3[SLO histograms]
    C4[audit counters bounded]
    C5[ladder-outcome aggregate]

```

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C6[lint exit codes]
end
C1 --> KA[K-ADOPT]
C2 --> KA
C3 --> KR[K-REL]
C4 --> KP[K-PRIV]
C5 --> KC[K-CENS]
C6 --> KP
KA --> B{privacy boundary:<br/>no forbidden label<br/>no per-
user flow}
KR --> B
KP --> B
KC --> B
B --> OUT[every KPI is counts-only · the absence is the promise]

```

Closing honesty (§11.4.6). No target in §3–§7 is a measured result; each is a business TARGET marked UNVERIFIED and met only when its named counts-only series produces data. And by design, no KPI anywhere in this document can be computed from a per-user flow, a destination, or a timestamp-of-use — because that data never exists (P7 / S6). The measurement boundary *is* the product.

Sources verified

- docs/research/mvp/final/00-product-scope-and-principles.md — principle P7 (no-logging by construction §8), parity F13/F14/F15 (§9), differentiators X1/X3 (§9), the 8-criteria MVP DoD (§10.2).
- docs/research/mvp/final/v03-control-plane/svc-telemetry.md §3 — the complete counts-only Prometheus registry (helix_reconcile_seconds, helix_revoke_enforce_seconds, helix_events_dlq_total, helix_stream_pending_entries, helix_presence_online_devices, helix_audit_events_total{action}, helix_build_info, process_resident_memory_bytes); §3.1 the forbidden-label cardinality bound (no tenant_id/device_id/ip label); §5 ephemeral presence; the per-tenant /v1/stats vs public /metrics split.
- docs/research/mvp/final/v03-control-plane/svc-coordinator.md §7 — the convergence < 1 s and revoke < 1 s SLO histograms feeding K-REL.
- docs/research/mvp/final/v05-security/no-logging-as-code.md — the stored-vs-never-stored partition (§2), coarse last_seen_at (§2.2), the schema-lint + payload-lint + meta-shape guard + paired §1.1 mutation feeding K-PRIV, the deliberately-absent data classes (§8) and the structural-impossibility framing.
- docs/research/mvp/final/v02-data-plane/transport-selection-ladder.md — the aggregate, no-per-user ladder telemetry (I5) and per-network fingerprint-hash memory feeding K-CENS.

Constitution bindings applied: §11.4.44 (revision header), §11.4.6 (no-guessing — every KPI target marked UNVERIFIED, never asserted as measured), §11.4.69 (KPIs derived only from captured counts-only series), §11.4.112 (the deliberately-absent metrics framed as structural impossibilities, not backlog), the no-logging covenant (P7 / S6) bounding what any KPI may measure, §11.4.65/§11.4.153 (HTML+PDF[+DOCX] exports follow in refinement).